Paleomagnetism of the Clam Bank Formation and paleolatitude estimates for western Newfoundland

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Received May 25, 1992
Revision accepted January 28, 1993

The Pridolian Clam Bank Formation around Lourdes Cove on the Port au Port Peninsula, western Newfoundland, underwent deformation during the Acadian orogeny. As a result, some of the beds were overturned, but the stratification planes can be accurately determined everywhere. Paleomagnetic studies of the Clam Bank Formation have yielded three well-defined components of magnetization, all acquired subsequent to the deformation event: component *A* with D = 337.3°, *I* = −28.3°, (*N* = 16 sites, *k* = 25.3, α_{95} = 7.5°), with a corresponding paleopole at 23.2°N, 145.0°E (dp, dm = 4.5°, 8.2°); component *B* with *D* = 172.9°, *I* = 5.7° (*N* = 35 specimens, *k* = 10.2, α_{95} = 6.4°), with a corresponding paleopole at 38.2°N, 130.1°E (dp, dm = 3.2°, 6.4°); component *C* with *D* = 350.4°, *I* = 69.8° (*N* = 33 specimens, *k* = 8.9, α_{95} = 8.9°). A pre-Mesozoic origin of the *A* and *B* components is indicated by the presence of normal and reversed components in specific sites; by the lack of correspondence between the *A* and *B* paleopoles and the Mesozoic and later pole positions from the Appalachians and the North American craton; and by agreement with Paleozoic poles from the region. The *A* component was probably acquired immediately after deformation during the Acadian orogeny. The *B* component is probably a chemical remanence that was acquired during Permo-Carboniferous (Kiaman) time. The *C* component is of recent origin, probably acquired in the present Earth's field. Paleomagnetic data from western Newfoundland are used in a localized setting to construct a paleopole sequence and to estimate paleolatitudes for western Newfoundland during the Paleozoic. Keeping in mind the paucity of data for Siluro-Devonian age from this region, western Newfoundland seems to have been at its southernmost position at the end of the Ordovician and to have occupied equatorial latitudes during the Permo-Carboniferous. The paleolatitude trend suggests that this block, which is part of the North American craton, moved in a southerly direction during the early Paleozoic and in a northerly direction during the middle and late Paleozoic.

La Formation de Clam Bank, d'âge pridolien, des environs de Lourdes Cove sur la péninsule de Port au Port, dans la partie occidentale de Terre-Neuve, fut déformée durant l'orogenèse de l'Acadien. Une conséquence de cette déformation est le renversement de certains lits, cependant les plans de stratification peuvent être déterminés précisément en tous lieux. Les études paléomagnétiques de la Formation de Clam Bank ont fourni trois composantes d'aimantation bien définies, toutes acquises subséquemment à l'événement de déformation : la composante *A* avec *D* = 337,3°, *I* = −28,3° (*N* = 16 sites, *k* = 25,3, α_{95} = 7,5°) avec son paléopôle correspondant à 23,2°N, 145,0°E (dp, dm = 4,5°, 8,2°); la composante *B* avec *D* = 172,9°, *I* = 5,7° (*N* = 35 spécimens, *k* = 10,2, α_{95} = 6,4°) avec son paléopôle correspondant à 38,2°N, 130,1°E (dp, dm = 3,2°, 6,4°); la composante *C* avec *D* = 350,4°, *I* = 69,8° (*N* = 33 spécimens, *k* = 8,9, α_{95} = 8,9°). L'acquisition antémésozoïque des composantes *A* et *B* est démontrée par la présence de composantes normales et inverses à des sites spécifiques; par l'absence au Mésozoïque de toute correspondance avec les paléopôles *A* et *B* et par les positions des pôles d'origine plus tardive dans les Appalaches et dans le craton de l'Amérique du Nord; et par une concordance avec les pôles paléozoïques de la région. La composante *A* fut probablement acquise immédiatement après la déformation durant l'orogenèse de l'Acadien. La composante *B* correspond probablement à une rémanence chimique qui fut acquise durant le Permo-Carbonifère (Kiamien). La composante *C* a une origine récente, probablement acquise du champ Terrestre actuel. Les données paléomagnétiques de la partie occidentale de Terre-Neuve sont utilisées dans un contexte localisé pour reconstituer la séquence des paléopôles et pour estimer les paléolatitudes durant le Paléozoïque de la partie occidentale de Terre-Neuve. Conscient du peu de données disponibles sur l'âge siluro-dévonien dans cette région, il semble tout de même que c'est à la fin de l'Ordovicien que la partie occidentale de Terre-Neuve eut atteint sa position la plus méridionale et que durant le Permo-Carbonifère elle occupait des latitudes équatoriales. Les mouvements des paléolatitudes suggèrent que ce bloc, qui représente une portion du craton de l'Amérique du Nord, s'est déplacé vers le sud durant le Paléozoïque précoce et vers le nord durant le Paléozoïque moyen et tardif.

[Traduit par la rédaction]

Can. J. Earth Sci. **30**, 776–786 (1993)

Objectives

Paleomagnetic studies have been carried out on red sandstone beds exposed on either side of Lourdes Cove, Port au Port peninsula, in western Newfoundland. The objectives of the studies were threefold. The first objective was to contribute to the data base for western Newfoundland that provides evidence of various motions involved in the closing of the Iapetus Ocean during the Paleozoic. The geology of the island of Newfoundland has been interpreted over the past 25 years as resulting from the opening and closing of an early Paleozoic Iapetus Ocean (Wilson 1966; Williams et al. 1972; Williams 1979 and references therein; McKerrow and Scotese 1990; Williams 1992). Apart from the rotations of western Newfoundland relative to cratonic North America that were suggested by Black (1964), western Newfoundland has always been considered part of cratonic North America. A comparison of western Newfoundland paleomagnetic data with data of contemporaneous rocks from Avalonia would provide reasonable constraints on the suggested motions at the time of the closing of the Iapetus Ocean.

The second objective was to establish whether the red sandstone units on either side of Lourdes Cove (Fig. 1) belong to the same formation. In most of the literature (Riley 1962; Rodgers and Neale 1963; Rodgers 1965; Fahraeus 1973; Williams 1985), these beds were recognized as part of the Clam Bank Formation. However, O'Brien (1975) argued that beds north of Lourdes Cove and extending northeast along the coast to Misty Point should be regarded as different in age

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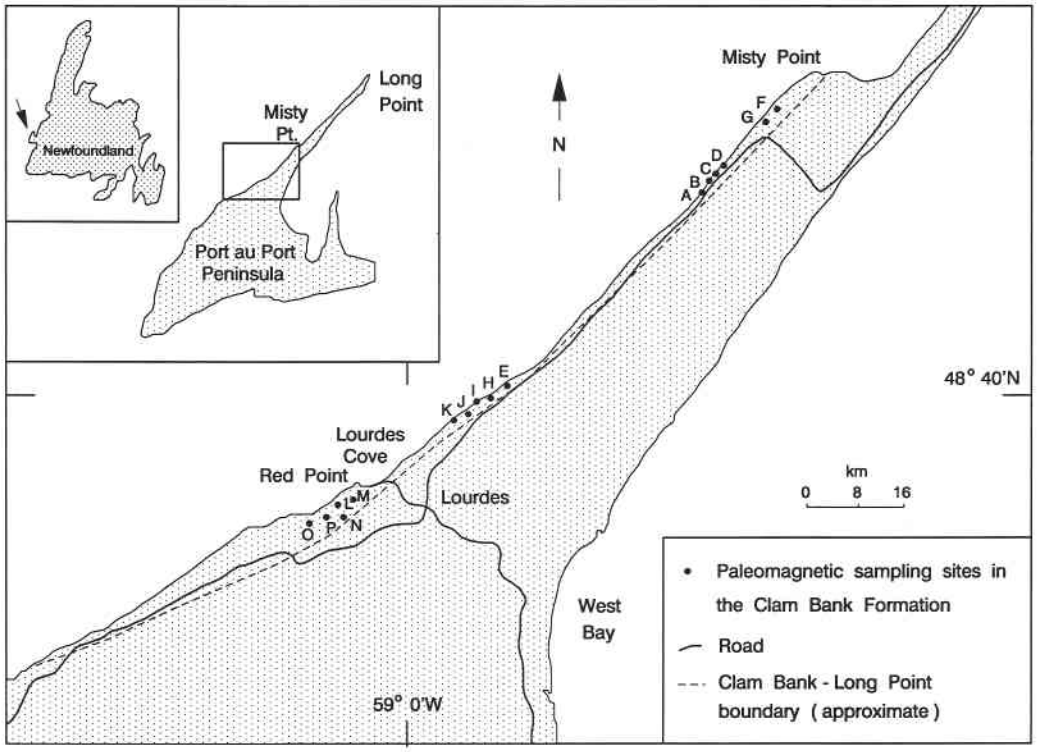


FIG. 1. Sketch map of the western part of the Port au Port Peninsula showing the Clam Bank Formation (simplified map from O'Brien 1975) and paleomagnetic sampling sites. Insets show the Port au Port Peninsula and the island of Newfoundland.

from the beds south of Lourdes Cove. She considered that beds north of Lourdes Cove are part of the Middle Ordovician Long Point Formation and that beds south of Lourdes Cove are part of the Pridolian Clam Bank Formation. Systematic paleomagnetic studies of these red beds, both north and south of Lourdes Cove, should provide convincing evidence of whether they are contemporaneous.

The third objective of the study was to attempt to draw an apparent polar wander path and to estimate paleolatitudes for western Newfoundland during the Paleozoic; this is possible if characteristic remanence directions, each with an associated certainty of age, can be established for these rocks. Since an adequate number of paleomagnetic results with reasonable age control is now available, such an attempt may be successful.

Geology and sampling

Williams (1979) summarized the geology of the island of Newfoundland in the broader context of the tectonics of the Appalachian orogen. Four distinct lithotectonic zones were recognized; from west to east, these were the Humber, Dunnage, Gander, and Avalon zones. The Port au Port Peninsula, in the Humber Zone, has been extensively studied geologically (Schuchert and Dunbar 1934; Sullivan 1940; Riley 1962; Rodgers and Neale 1963; Rodgers 1965; Fahraeus 1973; O'Brien 1975; Dean 1979; Williams 1985).

The rocks of interest in the present study are the coastal exposures of red sandstones between Misty Point and Red Point on either side of Lourdes Cove (Fig. 1). In most of the literature cited above, they are described as part of the Clam Bank Formation. They comprise mostly brick-red sandstone and mudstone. The sandstones exhibit cross-bedding and are commonly coarse grained. Units of fine-grained sandstone and siltstone alternate with mudstone and coarse-grained sandstone. At the north end of Lourdes Cove, rocks of the Clam Bank Formation conformably overlie the grey sandstones of the Middle Ordovician Long Point Group (Rodgers 1965; Williams 1985). At the south end of Lourdes Cove, Rodgers (1965) recognized Siluro-Devonian brachiopods in grey shales and sandstones of the Clam Bank Formation. Along Long Point, the Clam Bank Formation and the upper member of the Long Point Group are homoclinal. Strata of the two formations strike at approximately 240° and dip at 40–50° NW. From about 2 km northeast of Lourdes to Red Point, all exposures dip southeast, sometimes by as much as 80°. Rodgers (1965) recognized that the beds pass from one dip direction (northwest) to the other (southeast) through the vertical and, based on cross-bedding and other evidence, interpreted the southeast dipping beds as overturned, with all strata being young to the northwest. Table 1 lists the strikes and dips observed at each of the paleomagnetic collecting sites. The Clam Bank Formation was assigned an age (Rodgers 1965) close to the Silurian–Devonian transition. The timing of the overturning of the beds can be inferred from structural evidence.

The fact that sections of the upper members of the Long Point Group and the Clam Bank Formation were overturned, while the Carboniferous beds in the vicinity at Red Point and Red Island do not show any evidence of overturning, suggested to Rodgers that the overturning was a result of the Acadian orogeny. An alternate interpretation of the structure of these red beds was provided by O'Brien (1975). She suggested that the beds north of Lourdes Cove to Misty Point be regarded as the uppermost member of the Long Point Group, named it the Lourdes Member and assigned an age of uppermost Middle Ordovician to this member. The red sandstones and silt-

TABLE 1. *A*, *B*, and *C* magnetizations from the Clam Bank Formation

				A magnetization									
				Natural remanent magnetism				Demagnetized in situ				Demagnetized t.c.	
Site	Samples	Strike (°)/tilt (°)[a]	*N*	*D* (°)	*I* (°)	*k*	α_{95} (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	*D* (°)	*I* (°)
A	C44–C48	221/41	6	317.9	8.0	25.2	13.6	318.0	−21.2	16.4	11.4	325.4	−61.9
B	C49–C52	221/41	4	333.0	1.0	33.1	16.2	327.7	−22.0	44.7	8.9	343.9	−59.7
C	C53–C56	221/41	4	338.7	16.9	11.3	28.5	336.1	−16.1	138.5	7.2	352.4	−51.1
D	C57–C61	221/41	3	344.8	39.1	50.8	17.5	340.1	−20.4	37.9	13.8	0.0	−52.1
E	C62–C67	225/109	5	164.0	58.7	4.5	40.7	158.2	35.0	16.0	15.6	293.2	31.4
F	C68–C73	220/47	5	328.9	15.5	91.0	8.1	324.5	−19.0	56.7	6.9	342.1	−62.9
G	C74–C79	225/42	5	349.3	28.7	17.4	18.9	346.9	− 9.0	72.5	6.8	0.3	−42.5
H	C80–C85	238/107	5	113.3	74.1	17.0	19.1	157.0	20.0	20.3	12.1	313.0	50.1
I	C86–C91	238/107	4	85.0	83.9	89.4	9.8	136.3	43.4	13.6	23.7	336.6	28.1
J	C92–C97	237/104	6	181.5	15.1	3.2	45.1	153.0	39.8	12.5	15.6	320.5	35.2
K	C98–C104	240/102	5	133.5	70.7	2.8	56.7	161.9	21.0	33.0	8.4	309.5	54.7
L	D1–D7	252/120	12	355.4	−30.6	2.5	31.8	352.2	−27.9	10.0	8.3	149.2	−30.9
M	D8–D12	254/113	10	359.0	−18.2	34.4	8.4	356.5	−21.7	10.2	9.8	144.9	−42.9
N	D13–D16	248/123	8	332.3	−30.4	10.8	17.6	351.1	−42.9	8.9	11.3	147.1	−14.2
O	D17–D20	249/137	7	351.0	−57.1	16.1	14.2	355.2	−53.3	9.4	12.9	150.6	12.6
P	D21–D24	248/137	8	330.1	−47.1	17.0	13.8	330.3	−38.9	5.2	18.5	164.6	−4.7
Mean *A*	In situ		N = 16 sites			D = 337.3°, I = −28.3°, k = 25.3, α_{95} = 7.5°							
			N = 83 specimens			D = 337.1°, I = −31.5°, k = 20.2, α_{95} = 3.5°							
			Paleopole is at 23.2°N, 145.0°E, dp, dm = 4.5°, 8.2°										
	Relative to bedding		N = 16 sites			D = 330.7°, I = −54.3°, k = 1.2, α_{95} = 74.1°							
Mean *B*	In situ		N = 35 specimens			D = 172.9°, I = 5.7°, k = 10.2, α_{95} = 6.4°							
	Relative to bedding		N = 35 specimens			D = 248.9°, I = 64.8°, k = 3.7, α_{95} = 13.1°							
			Paleopole is at 38.2°N, 130.1°E, dp, dm = 3.2°, 6.4°										
Mean *C*	In situ		N = 33 specimens			D = 350.4°, I = 69.8°, k = 8.9, α_{95} = 8.9°							

NOTES: *N*, number of specimens used in demagnetization unless specified otherwise; *D*, declination; *I*, inclination; *k*, Fisher's precision parameter; α_{95}, radius of the 95% circle of confidence around the mean; in situ, relative to the present horizontal; t.c., relative to the bedding plane; dp, dm, size of the oval of confidence around the pole. Reverse *A* site and specimen directions, the normal *B* specimen directions, and the 2 reverse *C* directions have been inverted to calculate the respective mean directions.

[a]Tilt of >90° means an overturned bed.

stones south of Lourdes Cove were considered by O'Brien to be part of the Clam Bank Formation of Pridolian age. More recent papers on the geology of the area, however, follow Rodgers interpretation (e.g., Williams 1985) and consider that the Clam Bank Formation extends both northeast and southwest of Lourdes Cove.

A total of 84 oriented samples were collected from 16 sites of the Clam Bank Formation. The site localities are shown in Fig. 1. Careful repeated observations were made of the strike and dip at each site. The orientation of the samples was measured with a Brunton or Silva compass, verified occasionally by a solar compass. Block samples, as well as in situ drill cores, were taken. The collection includes a 24-sample section (D1 to D24) through a siltstone exposure south of Lourdes Cove where the dips are to the southeast. Varying tilt angles at different sites gives an opportunity to determine with reasonable confidence whether the isolated magnetization is pretilting or posttilting.

Paleomagnetism

Remanence measurements were made using a Schonstedt SSM-1 spinner magnetometer. Thermal demagnetizations were conducted using a Schonstedt TSD-1 thermal specimen demagnetizer where the ambient fields were less than 5×10^{-9} T. The natural remanent magnetism (NRM) intensities are weak (typical values $\sim 5.0 \times 10^{-4}$ A/m) but can be readily measured. At least one specimen (cylinder) from each independently oriented block or core (sample) was used in demagnetization. The NRM directions for specimens and site means are plotted in Fig. 2. The specimen magnetizations, while in general concentrated in northwesterly upward and southeasterly downward clusters, also have a number of steep directions. The majority of site means have northwesterly declinations with both up and down directions. A group of sites (north of Lourdes Cove and overturned) have mean directions with steep positive inclinations.

Each specimen was subjected to stepwise thermal demagnetization up to 680°C at close steps. Recognition of various components of magnetization was achieved through the use of one or more of the following analytical techniques: visual observation of vector diagrams (Zijderveld 1975) and their processing using principal component analyses (Kirschvink 1980); remagnetization circles (Halls 1976); and vector subtraction procedures (Roy and Park 1974; Hoffman and Day 1978).

The response of specimens to demagnetization is presented in Figs. 3–5. Specimens D1, D2.2, and D5.2 (site L) are from an overturned siltstone section and their behaviour is shown in Fig. 3. A northwesterly upward magnetization was removed at low and intermediate temperatures. A southerly, shallow magnetization was recognized between 600 and 680°C.

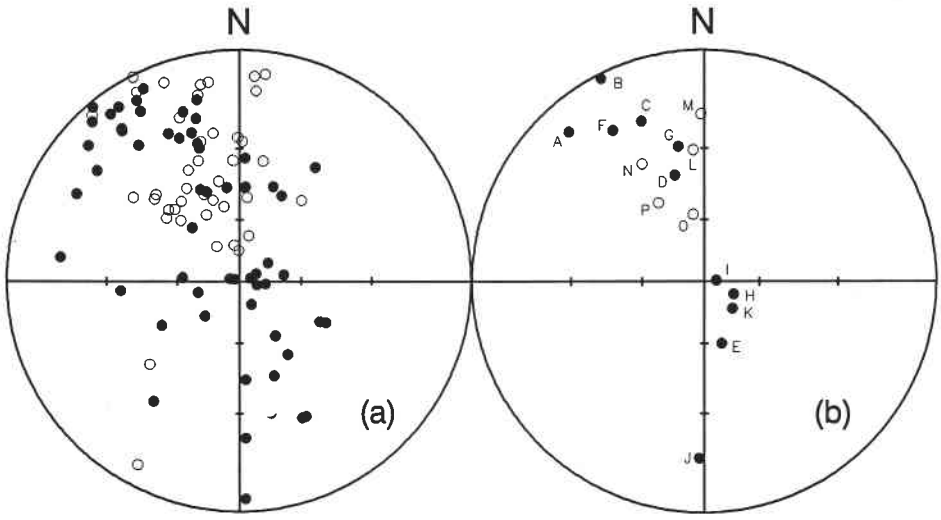


FIG. 2. Natural remanent magnetism (NRM) directions of the Clam Bank Formation plotted on a Wulff's net with respect to the present horizontal. (*a*) Specimen directions. (*b*) Site mean directions. Solid symbols indicate positive inclinations; open symbols indicate negative inclinations.

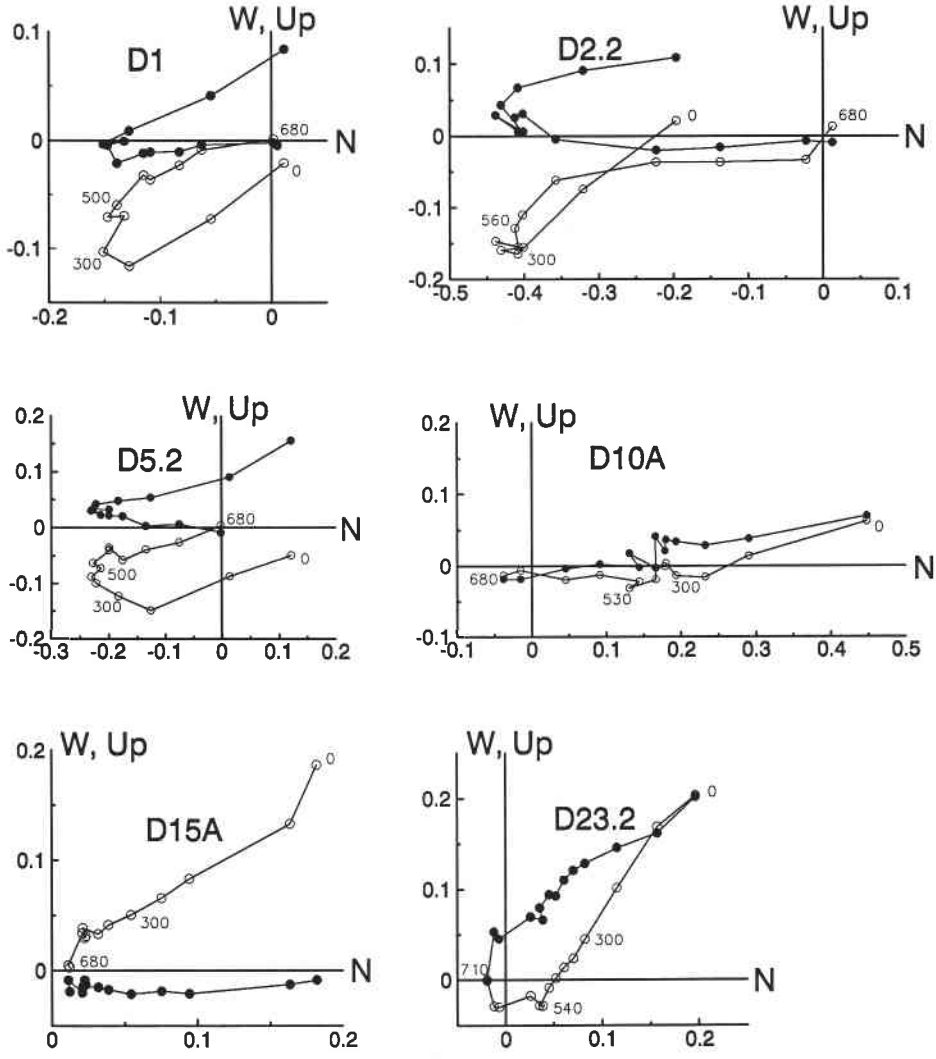


FIG. 3. Vector (Zijderveld) diagrams showing response of specimens to thermal demagnetization. Units are in mA/m. Solid symbols are projections in the horizontal plane and open symbols are projections in the north–south vertical plane. Numbers next to selected points on the vertical projection are the values of the temperature steps during thermal demagnetization.

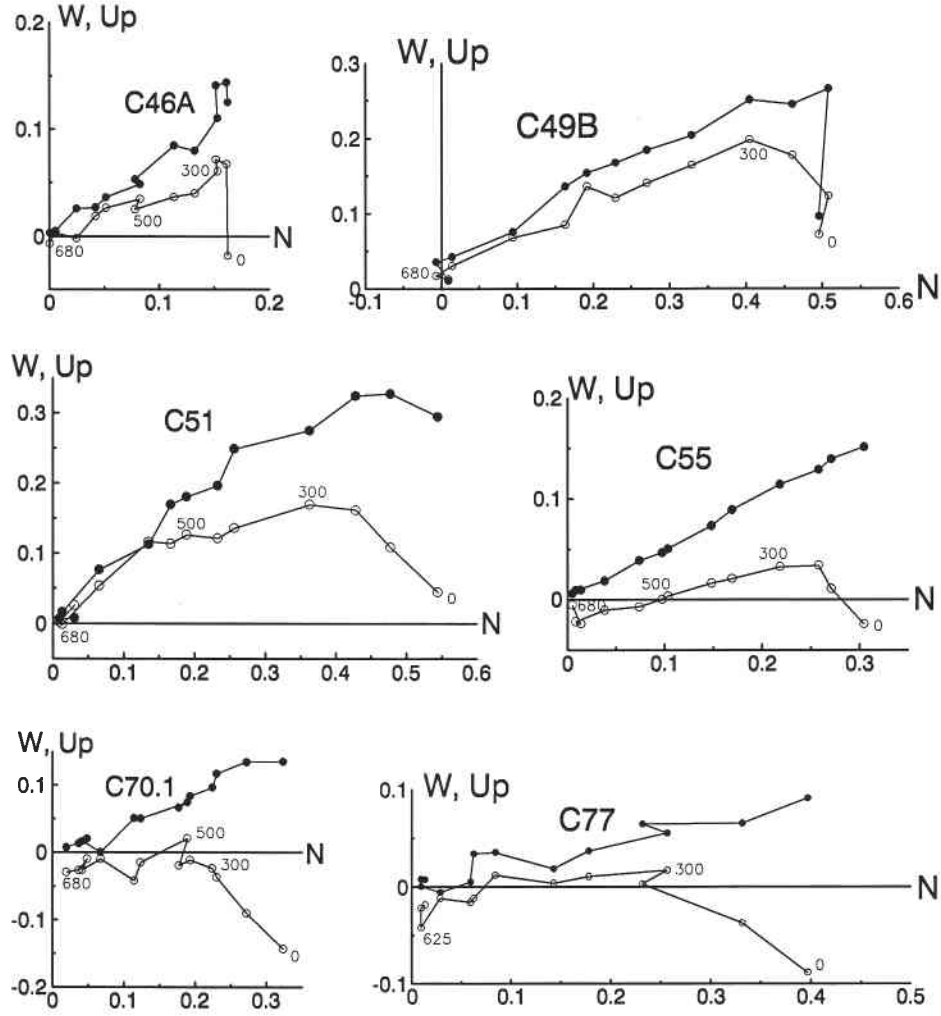


FIG. 4. Vector (Zijderveld) diagrams showing response of specimens to thermal demagnetization. Symbols and units are the same as in Fig. 3.

There seems to be an overlap in unblocking temperatures between the two components, as shown by the "shoulder" in the vector diagrams for these specimens between 300 and 500°C. In spite of this possible overlap in unblocking temperatures, the two components (the northwesterly and upward magnetization, hereinafter called *A* component, and the southerly and shallow magnetization, hereinafter called *B* component) can easily be recognized.

The behaviour of three specimens from three other nearby sites comprising overturned siltstone beds is also shown in Fig. 3. Specimen D10A shows the *A* component at lower blocking temperatures, followed by a northerly, and shallow magnetization (opposite to *B*) from 300 to 630°C. At temperatures beyond 660°C, the magnetization of this specimen moved to a southerly and shallow direction (*B* component). For specimens D15A and D23.2, a northwest and upward magnetization was removed during the initial stages of demagnetization, which is the same as the *A* component recognized earlier for other specimens. During the last stages of demagnetization, both specimens D15A and D23.2 showed a northerly and shallow magnetization (opposite of *B*). Specimen D23.2 moved to a southerly and shallow magnetization during the last step of demagnetization (*B* component).

Figure 4 shows the behaviour of two specimens from beds near Misty Point (site B.). Specimen C49B shows a steep downward-directed magnetization, probably representative of the present Earth's field (hereinafter called component *C*), at lower temperatures, and a characteristic magnetization northwesterly and upwards at higher temperatures. This characteristic remanence seems to be the same as the *A* component discussed earlier. Specimen C51, from the same area, also shows a downward-directed magnetization (*C*) followed by the characteristic *A* component. These specimens do not seem to carry the shallow magnetization with the north−south axis (*B* component).

The behaviour of four other specimens with a similar pattern is shown in Fig. 4. Following removal of the *C* magnetization at lower unblocking temperatures, the directions correspond to the *A* magnetization. Typical behaviour of specimens from the overturned beds around Lourdes Cove (sites H, J, and K) is shown in Fig. 5. Specimen C82B, from one of the overturned beds, exhibits three components. A steep downward-directed magnetization (*C*) is followed by an inverted *B* and the previously recognized *B* magnetizations. Specimen C96.1 shows the *C* component, followed by a southeast-directed downward magnetization, which is interpreted as a reverse *A* magnetization. In some cases, *C* is followed by the inverted *B* component (C83.2). The magnetization of each specimen was individually analyzed and the various components isolated in the study are shown in Figs. 6−8.

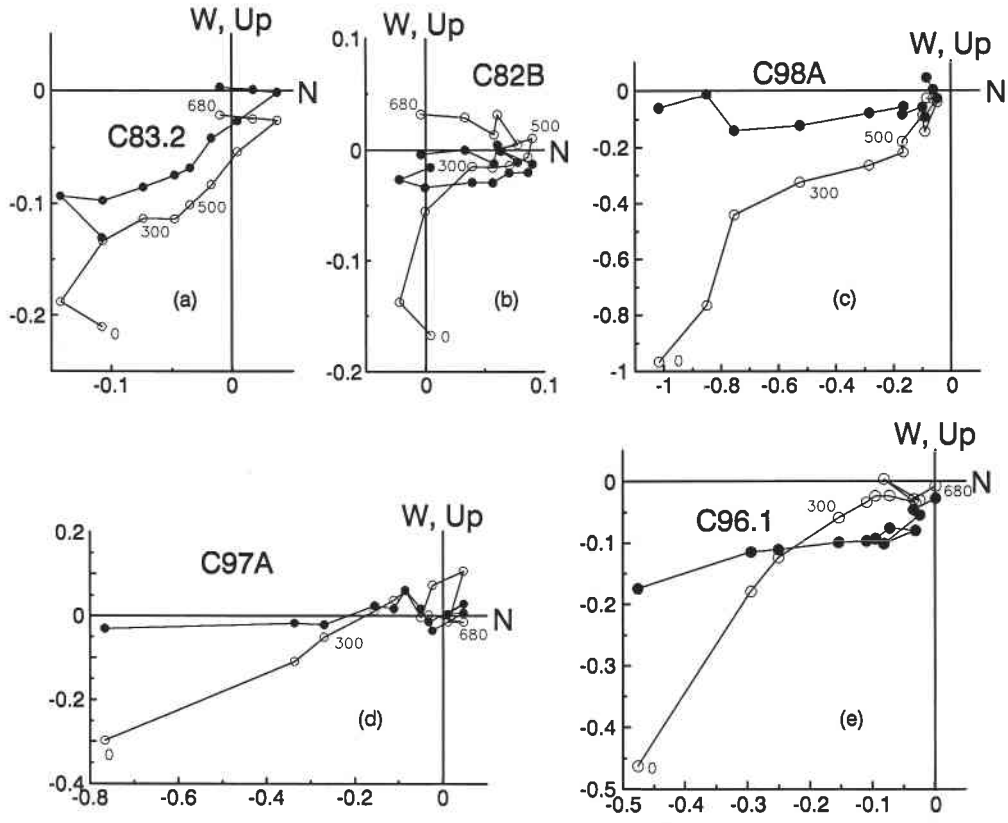


FIG. 5. Vector (Zijderveld) diagrams showing response of specimens to thermal demagnetization. Symbols and units are the same as in Fig. 3.

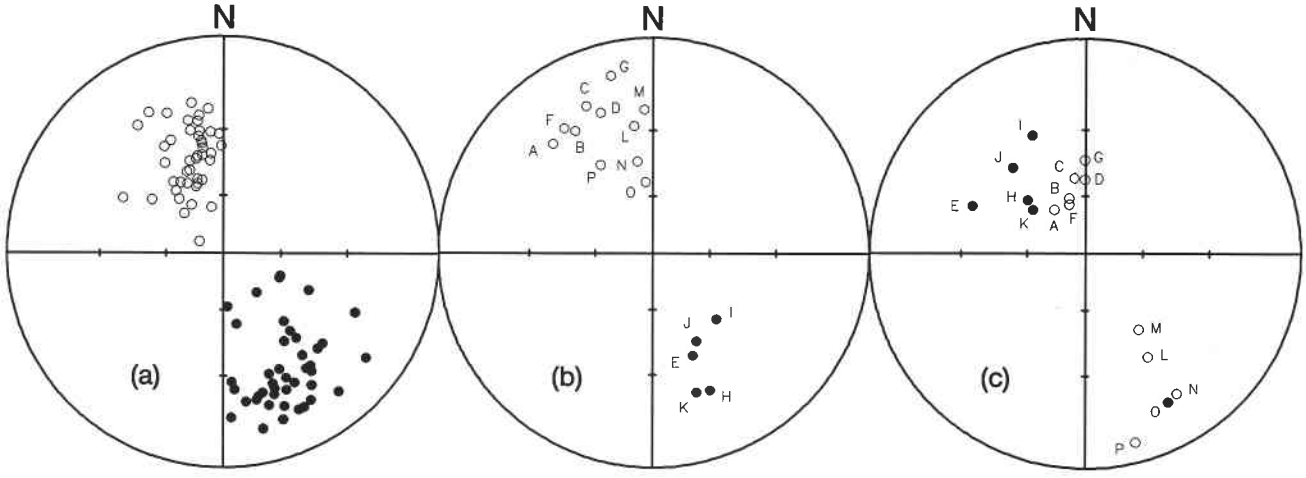


FIG. 6. Wulff's net showing *A* magnetization isolated from the Clam Bank Formation. Solid symbols indicate positive inclinations; open symbols indicate negative inclinations. (*a*) Individual specimens showing the *A* magnetization relative to the present horizontal. (*b*) Site mean directions relative to the present horizontal. (*c*) Site mean directions relative to the bedding plane.

Discussion

The following magnetizations were recognized for samples of the Clam Bank Formation: *A* normal and *A* reversed; *B* normal and *B* reversed; and *C*. The presence of reversals gives confidence that the *A* and *B* magnetizations are paleomagnetic in nature and are not artifacts of the analyses. Different structural corrections at different sites allow inferences to be made about the age of acquisition of the *A* and *B* components. This is equivalent to comparison of the respective values of Fisher's precision parameter using a fold test (McElhinny 1964; McFadden and Jones 1981). Comparison of data in situ and relative to bedding (Fig. 6 and Table 1 for the *A* magnetization and Fig. 7 and Table 1 for the *B* magnetization) clearly shows that both *A* and *B* were acquired after the overturning of the beds. Some factors that can aid in assigning ages to *A* and *B* magnetizations are the following: *A* normal is found in all the sites, while *A* reversed is confined only to overturned beds. The *B* magnetization is found only in the overturned beds.

A *component*

The *A* component is bipolar, with a northwest–southeast axis and intermediate upward–downward inclination, as seen both at specimen and site level (Figs. 6*a*, 6*b*; Table 1). It

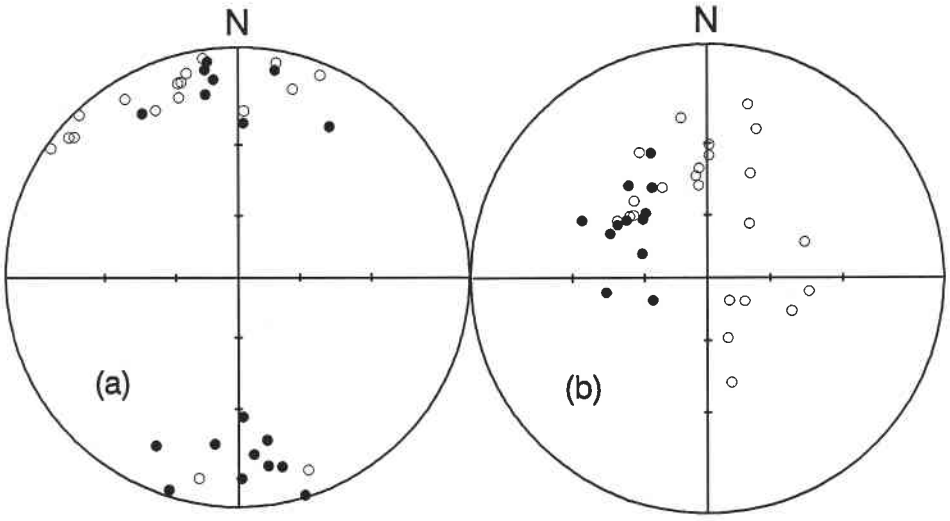


FIG. 7. Wulff's nets showing *B* magnetization isolated from individual specimens of the Clam Bank Formation. Solid symbols indicate positive inclinations; open symbols indicate negative inclinations. (*a*) Magnetization is relative to the present horizontal. (*b*) Magnetization is relative to the bedding plane.

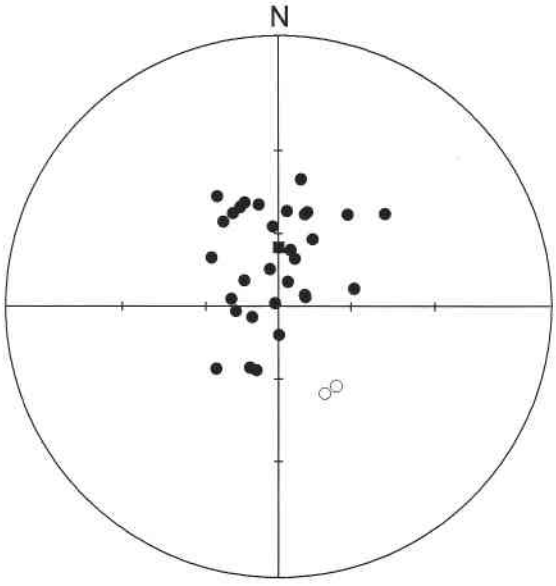


FIG. 8. Wulff's net showing *C* magnetization isolated from individual specimens of the Clam Bank Formation. Solid circles indicate positive inclinations; open circles indicate negative inclinations. The directions are relative to the present horizontal. ■ indicates the present-day Earth's dipole field.

should be noted here that Black (1964) reported a mean direction of $D = 337.5°$, $I = -19.8°$ ($N = 18$ samples, $k = 10.2$, $\alpha_{95} = 11.4°$) that is not significantly different from the in situ direction quoted in Table 1 for the *A* component. However, Black's results are at the NRM level (although he noted that alternating field demagnetization up to 30 mT did not affect the directions significantly), and he did not attempt any structural corrections. In our result, the grouping became progressively worse when corrected stepwise for structure. Fisher's precision parameter falls to 1.2 on 100% structural correction (quoted in Table 1) and the grouping becomes random (Fig. 6*c*).

The beds suffered deformation and overturning during the Acadian orogeny. Normal *A* is found in all the sites and reverse *A* is present only in the overturned beds. It is suggested therefore that the acquisition of the *A* magnetization was probably associated with events of the Acadian orogeny. The duration of overturning was probably long enough to record the reverse *A* magnetization as well.

It should be noted, however, that when combined according to the sites, there is an apparent correlation between the site locations (Fig. 1) and the value of the *A* directions. Sites A, B, C, D, F, and G, which comprise the northwest-dipping beds (called group 1 sites in this discussion), show northwest declination and comparatively shallow, upward inclination. Sites E, H, I, J, and K, with southeast dips (overturned northwest dip), and north of Lourdes Cove (hereafter called group 2 sites), show southeast declination and intermediate downward inclination. Sites L, M, N, O, and P, with southeast dips (overturned northwest dip), and south of Lourdes Cove (hereafter called group 3 sites), show northwest declination and intermediate upward inclination. After structural correction (Fig. 6*c*), this apparent separation becomes more pronounced, with group 3 and group 2 site directions moving, respectively, to the southeast and up, and to the northwest and down, while the group 1 sites show north–northwesterly and comparatively steep upward directions.

If all the sites were treated as belonging to the same formation, Fisher's precision parameter would fall to 1.2 on structural correction, corresponding to a random grouping in Fig. 6*c*. However, if we consider the possibility that the northwest- and southeast-dipping (overturned) beds are different in age, then the following means can be obtained relative to the bedding plane. Northwest-dipping group 1 sites: $N = 6$ sites, $D/I = 349.2°/-55.6°$, $k/\alpha_{95} = 56.8/9.0°$ with paleopole at 4.7°N, 129.8°E. Southeast-dipping group 2 and group 3 sites: $N = 10$ sites, $D/I = 324.3°/28.9°$, $k/\alpha_{95} = 11.3/15.0°$ with paleopole at 45.8°N, 174.9°E. However, this possibility is rejected on the following grounds: (1) the direction and the paleopole obtained for group 2 and 3 sites do not correspond to those of Paleozoic age from the North American craton and are not meaningful; (2) although the paleopole quoted above for group 1 sites may be meaningful, there is no reason (based on the magnetic behaviour of the samples) to

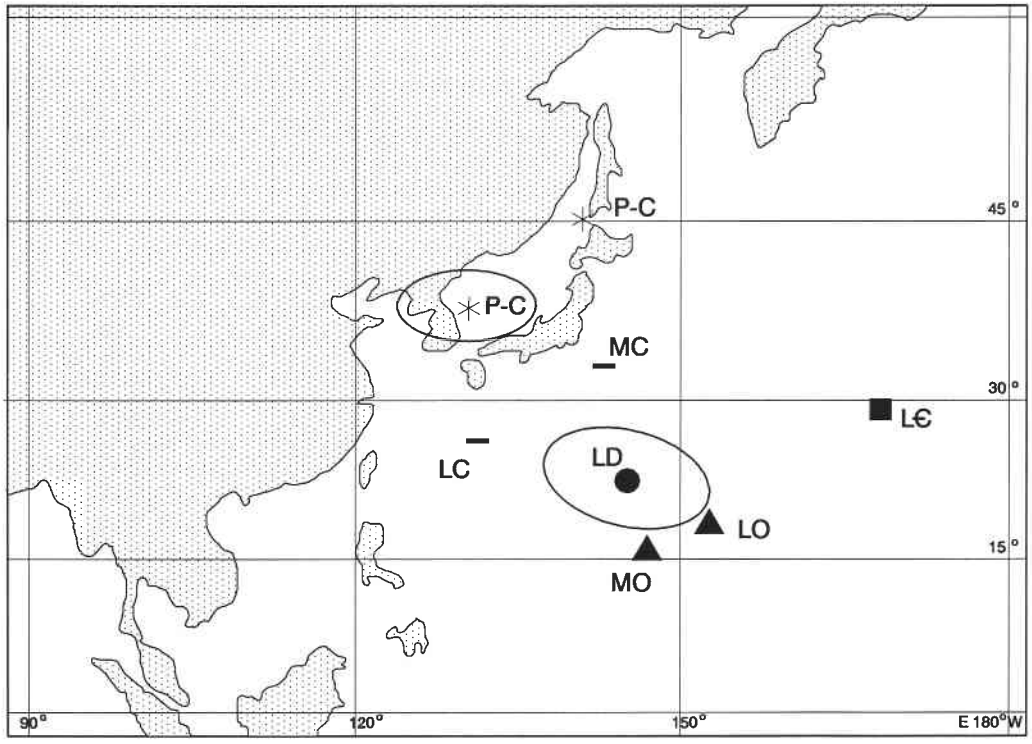


FIG. 9. Mean paleopoles relative to western Newfoundland from data with $Q > 2$ listed in Van der Voo (1990) and from the present study. ■ L-Є, Lower Cambrian, based on Bradore sandstone result (Rao and Deutsch 1976); ▲ LO, Lower Ordovician, based on St. George Group limestone result (Deutsch and Prasad 1987); ▲ MO, Middle Ordovician, based on three results from the Table Head Group (Deutsch and Prasad 1987; Hall and Evans 1988; Hodych 1989) and the Steel Mountain result (Murthy and Rao 1976); ● LD, Lower Devonian, based on the Clam Bank *A* magnetization (present study); ▬ LC, Lower Carboniferous, based on three results (Deer Lake Formation, Irving and Strong 1984; Jeffreys Village Member, Murthy 1985; Spout Falls Formation, Murthy 1985); ▬ MC, Middle Carboniferous, based on Barachois Group result (Murthy 1985); ✳ P-C, Permo-Carboniferous. Of the two Permo-Carboniferous poles, the one with the 95% oval of confidence is based on the Clam Bank *B* magnetization (present study), and the one without the oval is based on the Deer Lake *H* magnetization (Irving and Strong 1984).

exclude the group 2 and 3 sites; (3) the groups 2 and 3 mean directions after structural correction are not identical at 95% probability according to an F-test; (4) considering northwest- and southeast-dipping beds as different geological formations is not consistent with earlier geological work in the area. O'Brien's (1975) classification into the Long Point and Clam Bank formations is not based on the way the beds are dipping, and her boundary between the formations is at Lourdes Cove, between the two southeast-dipping site groupings, and not between the group 1 and group 2 and 3 sites. Thus, we reject the grouping of the sites as given above. We suggest the apparent correlation between site groupings and the value of the *A* component in Fig. 6*c* is due to possible overestimates in the magnitude of the structural corrections. We conclude that the paleomagnetically meaningful *A* component is the one that is presented in Figs. 6*a* and 6*b*.

B *and* C *components*

The *B* and *C* magnetizations for individual specimens are shown in Figs. 7 and 8. The *B* magnetization is probably Kiaman (Permo-Carboniferous) and is of chemical origin. It was imprinted only in the overturned beds, which probably developed sufficient porosity during the Acadian events to allow fluid migration through them at a later time, resulting in the chemical remanence. The relatively shallow northwest-dipping beds north of Lourdes Cove were spared the Kiaman overprint, as fluid migration is less likely in beds that did not suffer overturning. In this explanation, the presence of both reverse and normal Kiaman magnetizations (corresponding, respectively, to southerly and shallow, and northerly and shallow *B* magnetizations) poses a problem, as Kiaman is predominantly a reversed magnetization. However, normal remagnetizations of Permo-Carboniferous age have been reported earlier in the literature (see Opdyke 1992). The *C* magnetization (Fig. 8) is probably of recent origin, corresponding to the present Earth's field direction. In two cases, an upward-directed *C* component was isolated, as shown in Fig. 8.

To infer the nature of the magnetic minerals, we studied the acquisition and decay of isothermal remanent magnetization (IRM), and also the variation of saturation magnetization with temperature. Other rock magnetic experiments, for example, to determine the variation of magnetic susceptibility with temperature and to measure hysteresis parameters (i.e., to study the bulk magnetic properties associated with magnetic susceptibility), were not successful because the samples are extremely weak magnetically. The IRM curves (not shown) indicate that although an apparent plateau is reached in a field of about 200 mT or less, the magnetic moment was found to increase even up to 860 mT, the maximum field available in our experimental setup. This suggests that both magnetite and hematite are present in the samples. The $J_s - T$ curves show a Curie temperature dominantly around 570°C and the samples become extremely weak beyond that temperature. Although this suggests the presence of magnetite, the possible coexis-

TABLE 2. Estimates of paleolatitudes for western Newfoundland

Age	Paleolatitude	Error
Permo-Carboniferous	5.6°N	±5.2°
Permo-Carboniferous (Clam Bank *B*)	3.0°S	±3.2°
Middle Carboniferous	5.1°S	±3.6°
Lower Carboniferous	14.9°S	±8.1°
Lower Devonian (Clam Bank *A*)	13.6°S	±4.5°
Middle Ordovician	21.9°S	±7.2°
Lower Ordovician	17.9°S	±5.8°
Lower Cambrian	2.2°S	±2.5°

NOTES: Except for the Clam Bank data themselves, a fictitious direction of magnetization was calculated for each study area after "relocating" it on Port au Port Peninsula (Clam Bank site). The paleolatitude of the Clam Bank site was calculated corresponding to this direction. The error quoted is either (*i*) the dp value for the result if only one result is used, or (*ii*) the dp value estimated for the mean of a number of results. For a description of the results used, see caption to Fig. 9.

tence of hematite with magnetite cannot be easily inferred from the J_s-T curves unless the proportion of hematite is considerable. It is concluded from these experiments that the remanence carriers are both magnetite and hematite, both magnetic minerals carrying both the *A* and *B* components.

Paleozoic paleopoles and paleolatitude estimates

Table 1 gives pole positions corresponding to the *A* and *B* magnetizations, which were shown earlier to correspond in ages to Early Devonian and Permo-Carboniferous (Kiaman), respectively. It has long been a problem to recognize lower and middle Paleozoic magnetizations from cratonic rocks, a difficulty that has been noted and discussed frequently (Roy 1982; Roy and Morris 1983; Irving and Strong 1984; Kent and Opdyke 1985; Van der Voo 1989; and others). Thus, there are very few results for the early and middle Paleozoic age for the North American craton (Van der Voo 1989, 1990). Van der Voo (1989, 1990) plotted apparent polar wander paths for Europe and North America and discussed the results on a global scale. On a much more local scale, an attempt is made here to compile paleopole positions for western Newfoundland alone.

Although western Newfoundland has always been considered part of cratonic North America, we feel justified in compiling data for this physically distinct block rather than for the entire North American craton. We do not have many new results to add to the above-mentioned discussions of the larger tectonic setting. On the other hand, a reasonable amount of data is now available for western Newfoundland, and a compilation of data for this physically distinct block is useful. Only western Newfoundland results are included here, keeping in mind the possible rotations of western Newfoundland hypothesized by Black (1964); although they were not substantiated by later studies (e.g., Deutsch and Rao 1977). Paleopoles from eastern and central Newfoundland are similarly excluded because of possible motions between western Newfoundland and the rest of Newfoundland during the closing of Iapetus. Thus, recently obtained paleopoles that undoubtedly pass Van der Voo's Q-test (e.g., Moreton's Harbour Group, Johnson et al. 1991; Robert's Arm volcanics, Van der Pluijm et al. 1990; King George IV Lake redbeds and volcanics, Buchan and Hodych 1989; Springdale Group, Potts et al. 1992; Lawrenceton Formation of the Botwood Group, Gales et al. 1989) are not included in this discussion. There are more paleopoles available for central and eastern Newfoundland than for the western region. Van der Voo (1990) cited 10 Paleozoic (Cambrian to Carboniferous) results for western Newfoundland that have a $Q > 2$. A Kiaman result from the Deer Lake Group (Irving and Strong 1984) can be added to the 10 quoted by Van der Voo. These results have been combined according to age and plotted in Fig. 9 along with our results from the Clam Bank Formation. One glaring drawback of Fig. 9 is the lack of results for the Silurian and Devonian, except for a single Devonian pole (Clam Bank Formation, this paper). Although discussions of paleopole sequences are meaningful only when compared with similar sequences from other tectonic blocks, it may be noted that the cusp at the end of the Ordovician may be interpreted as a change in the direction of motion of western Newfoundland. However, the cusp needs to be substantiated by more data for the middle Paleozoic from western Newfoundland.

A more meaningful parameter in deciphering the motion of a tectonic block is the variation of its paleolatitude with time. We have estimated some Paleozoic paleolatitudes for western Newfoundland, which are presented in Table 2. The quoted standard deviations are based either on the original data by the authors or, when data were combined, on the number of results. Consistent with the paleopole data, it may be noted that western Newfoundland seems to be at its southernmost latitude at the end of the Ordovician, and moved to equatorial latitudes during Kiaman times. However, there are large gaps in data for the middle Paleozoic, and more quantitative estimates of past latitude positions can only be made after filling in the gaps.

Conclusions

1. A paleomagnetic study of the Pridolian Clam Bank Formation from western Newfoundland has yielded three components of magnetization, termed *A*, *B*, and *C*. Both normal and reversed *A* and *B* magnetizations are recognized. Based on different structural corrections required for data at different sites, it is concluded that neither *A* nor *B* are primary, i.e., correlative with the Pridolian age of the Clam Bank Formation.

2. The *A* magnetization, with a northwest–southeast axis and intermediate upward–downward inclination, is probably of Early Devonian age and is associated with events related to overturning of the Clam Bank beds.

3. The *B* magnetization, with a north–south axis and almost zero inclination, is probably of Permo-Carboniferous (Kiaman) age. The presence of a normal magnetization *B* is additional evidence that, while the Kiaman field was predominantly reversed, normal magnetization episodes did occur during this superchron.

4. The *C* magnetization is steep downward and is probably of recent origin.

5. A lack of evidence for primary remanence in red beds on both sides of Lourdes Cove prevented us from saying unambiguously that they all belonged to the same formation. However, the existence of the same secondary magnetization in red beds both north and south of Lourdes Cove suggests that they all are older than the Acadian remagnetization episode.

6. Paleolatitudes and paleopole estimates for western Newfoundland suggest that this region, which is part of the North American craton, moved southward during the early Paleozoic, with a shift in the direction of motion probably at the end

of the Ordovician. The timing of the shift in the direction of motion can be better established, however, once paleomagnetic results for the latest Silurian – earliest Devonian from western Newfoundland become available. The region later moved northward and occupied equatorial latitudes during Permo-Carboniferous time.

Acknowledgments

Tom Vallis helped with the preparation of the samples and Susan Gower and Michael Tubrett did the bulk of the laboratory measurements. Comments and suggestions by our colleague Ernst Deutsch and by referees, Rob Van der Voo and Ken Buchan, led to many improvements in the paper. G.S.M. is grateful to Rob Van der Voo, Ernst Deutsch, and the late Jean Roy for many discussions on the subject matter of this paper. Funds from Natural Sciences and Engineering Research Council of Canada Operating Grants, and Energy, Mines and Resources Canada Research Agreements were helpful in conducting the field work and in the laboratory processing of the samples.

Black, R.F. 1964. Palaeomagnetic support of the theory of rotation of the western part of the island of Newfoundland. Nature (London), **202**: 945–948.

Buchan, K.L., and Hodych, J.P. 1989. Early Silurian paleopole for redbeds and volcanics of the King George IV Lake area, Newfoundland. Canadian Journal of Earth Sciences, **26**: 1904–1917.

Dean, W.T. 1979. Trilobites from the Long Point Group (Ordovician), Port au Port peninsula, southwestern Newfoundland. Geological Survey of Canada, Bulletin 290.

Deutsch, E.R., and Prasad, J.N. 1987. Ordovician paleomagnetic results from the St. George and Table Head carbonates of western Newfoundland. Canadian Journal of Earth Sciences, **24**: 1785–1796.

Deutsch, E.R., and Rao, K.V. 1977. New palaeomagnetic evidence fails to support rotation of western Newfoundland. Nature (London), **266**: 314–318.

Fahraeus, L.E. 1973. Depositional environments and conodont-based correlation of the Long Point Formation (Middle Ordovician), western Newfoundland. Canadian Journal of Earth Sciences, **10**: 1822–1833.

Gales, J.E., Van der Pluijm, B.A., and Van der Voo, R. 1989. Paleomagnetism of the Lawrenceton Formation volcanic rocks, Silurian Botwood Group, Change Islands, Newfoundland. Canadian Journal of Earth Sciences, **26**: 296–304.

Hall, S.A., and Evans, I. 1988. Paleomagnetic study of the Ordovician Table Head Group, Port au Port Peninsula, Newfoundland. Canadian Journal of Earth Sciences, **25**: 1407–1419.

Halls, H.C. 1976. A least squares method to find a remanence direction from converging remagnetization circles. Geophysical Journal of the Royal Astronomical Society, **45**: 297–304.

Hodych, J.P. 1989. Limestones of western Newfoundland that magnetized before Devonian folding but after Middle Ordovician lithification. Geophysical Research Letters, **16**: 93–96.

Hoffman, K.A., and Day, R. 1978. Separation of multi-componen NRM: a general method. Earth and Planetary Science Letters, **40**: 433–438.

Irving, E., and Strong, D.F. 1984. Palaeomagnetism of the Early Carboniferous Deer Lake Group, western Newfoundland: no evidence for mid-Carboniferous displacement of "Acadia." Earth and Planetary Science Letters, **69**: 379–390.

Johnson, R.J.E., Van der Pluijm, B.A., and Van der Voo, R. 1991. Paleomagnetism of the Moreton's Harbour Group, northeastern Newfoundland Appalachians: evidence for an Early Ordovician island arc near the Laurentian margin of Iapetus. Journal of Geophysical Research, B, **96**: 11 689 – 11 702.

Kent, D.V., and Opdyke, N.D. 1985. Multicomponent magnetizations from the Mississippian Mauch Chunk Formation of the central Appalachians and their tectonic implications. Journal of Geophysical Research, B, **90**: 5371–5383.

Kirschvink, J.L. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

McElhinny, M.W. 1964. Statistical significance of the fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **8**: 338–340.

McFadden, P.L., and Jones, D.L. 1981. The fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **67**: 53–58.

McKerrow, W.S., and Scotese, C.R. 1990. Palaeozoic palaeogeography and biogeography. Geological Society of London, Memoir 12.

Murthy, G.S. 1985. Paleomagnetism of certain constituents of the Bay St. George sub-basin, western Newfoundland. Physics of the Earth and Planetary Interiors, **39**: 89–107.

Murthy, G.S., and Rao, K.V. 1976. Paleomagnetism of Steel Mountain and Indian Head anorthosites from western Newfoundland. Canadian Journal of Earth Sciences, **13**: 75–83.

O'Brien, F.H.C. 1975. The stratigraphy and palaeontology of the Clam Bank Formation, and upper part of the Long Point Formation of the Port au Port peninsula on the west coast of Newfoundland. M.Sc. thesis, Memorial University of Newfoundland, St. John's.

Opdyke, N.D. 1992. The Kiaman revisited. EOS, Transactions of the American Geophysical Union, **73**: 52.

Potts, S., Van der Pluijm, B., and Van der Voo, R. 1992. Paleomagnetic investigation of the Silurian Springdale Group, northcentral Newfoundland. EOS, Transactions of the American Geophysical Union, **73**: 88.

Rao, K.V., and Deutsch, E.R. 1976. Paleomagnetism of the Lower Cambrian Bradore sandstones, and the rotation of Newfoundland. Tectonophysics, **33**: 337–357.

Riley, G.C. 1962. Stephenville map area, Newfoundland. Geological Survey of Canada, Memoir 323.

Rodgers, J. 1965. Long Point and Clam Bank Formations, western Newfoundland. Proceedings of the Geological Association of Canada, **16**: 83–94.

Rodgers, J., and Neale, E.R.W. 1963. Possible "Taconic" klippen in western Newfoundland. American Journal of Science, **261**: 713–730.

Roy, J.L. 1982. Paleomagnetism of Siluro-Devonian rocks from eastern Maine: Discussion. Canadian Journal of Earth Sciences, **19**: 225–232.

Roy, J.L., and Morris, W.A. 1983. A review of paleomagnetic results from the Carboniferous of North America: the concept of Carboniferous geomagnetic field horizon markers. Earth and Planetary Science Letters, **65**: 167–181.

Roy, J.L., and Park, J.K. 1974. The magnetization process of certain red beds: vector analysis of chemical and thermal results. Canadian Journal of Earth Sciences, **11**: 437–471.

Schuchert, C., and Dunbar, C.O. 1934. Stratigraphy of western Newfoundland. Geological Society of America, Memoir 1.

Sullivan, J.W. 1940. The geology and mineral resources of the Port au Port peninsula, Newfoundland. Ph.D. thesis, Yale University, New Haven, Conn.

Van der Pluijm, B., Johnson, R.J.E., and Van der Voo, R. 1990. Early Paleozoic paleogeography and accretionary history of the Newfoundland Appalachians. Geology, **18**: 898–901.

Van der Voo, R. 1989. Paleomagnetism of North America; the craton, its margins, and the Appalachian belt. *In* Geophysical framework of the continental United States. *Edited by* L.C. Pakiser. Geological Society of America, Memoir 172, pp. 447–470.

Van der Voo, R. 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions. Reviews of Geophysics and Space Physics, **28**: 167–206.

Williams, H. 1979. Appalachian orogen in Canada. Canadian Journal of Earth Sciences, **16**: 792–807.

Williams, H. 1985. Geology, Stephenville map-area, Newfoundland. Geological Survey of Canada, Map 1579A, scale 1 : 100 000.

Williams, H. 1992. Did the Atlantic close and then re-open?: a commentary. The Tuzo Wilson Cycle: A 25th Anniversary Symposium, Geological Association of Canada, Program and Abstracts, p. 7.

Williams, H., Kennedy, M.J., and Neale, E.R.W. 1972. The Appalachian Structural Province. *In* Variations in tectonic styles in Canada. *Edited by* R.A. Price and R.J.W. Douglas. Geological Association of Canada, Special Paper 11, pp. 181–261.

Wilson, J.T. 1966. Did the Atlantic close and then re-open? Nature (London), **211**: 676–681.

Zijderveld, J.D.A. 1975. Paleomagnetism of the Esterel rocks. Ph.D. thesis, State University of Utrecht, The Netherlands.